

Introduction

Red-Drop is a web-based blood bank management system that streamlines donor registration, eligibility screening, blood inventory tracking, and donor–patient matching. It enables hospitals, admins, donors, and patients to interact through a centralized platform, ensuring efficient, transparent, and timely blood management with real-time notifications and analytics.

Aims

The main aim of Red-Drop is to provide an efficient solution forblood bank management that simplifies the process of blood donation and enhances accessibility for patients in need. It seeks to improve transparency and coordination among donors, hospitals, and patients.

Objectives

- Donor Management: Registration, document upload, eligibility check, and donation tracking
- Blood Inventory: Real-time monitoring and management of blood stock
- Smart Matching: Connects patients with compatible nearby donors
- Document Verification: Admin-approved identity and medical validation
- Notifications: Email/SMS alerts for eligibility, schedules, and requests

Research Question

How can a digital blood bank system in Nepal improve efficiency, donor retention, and patient outcomes?

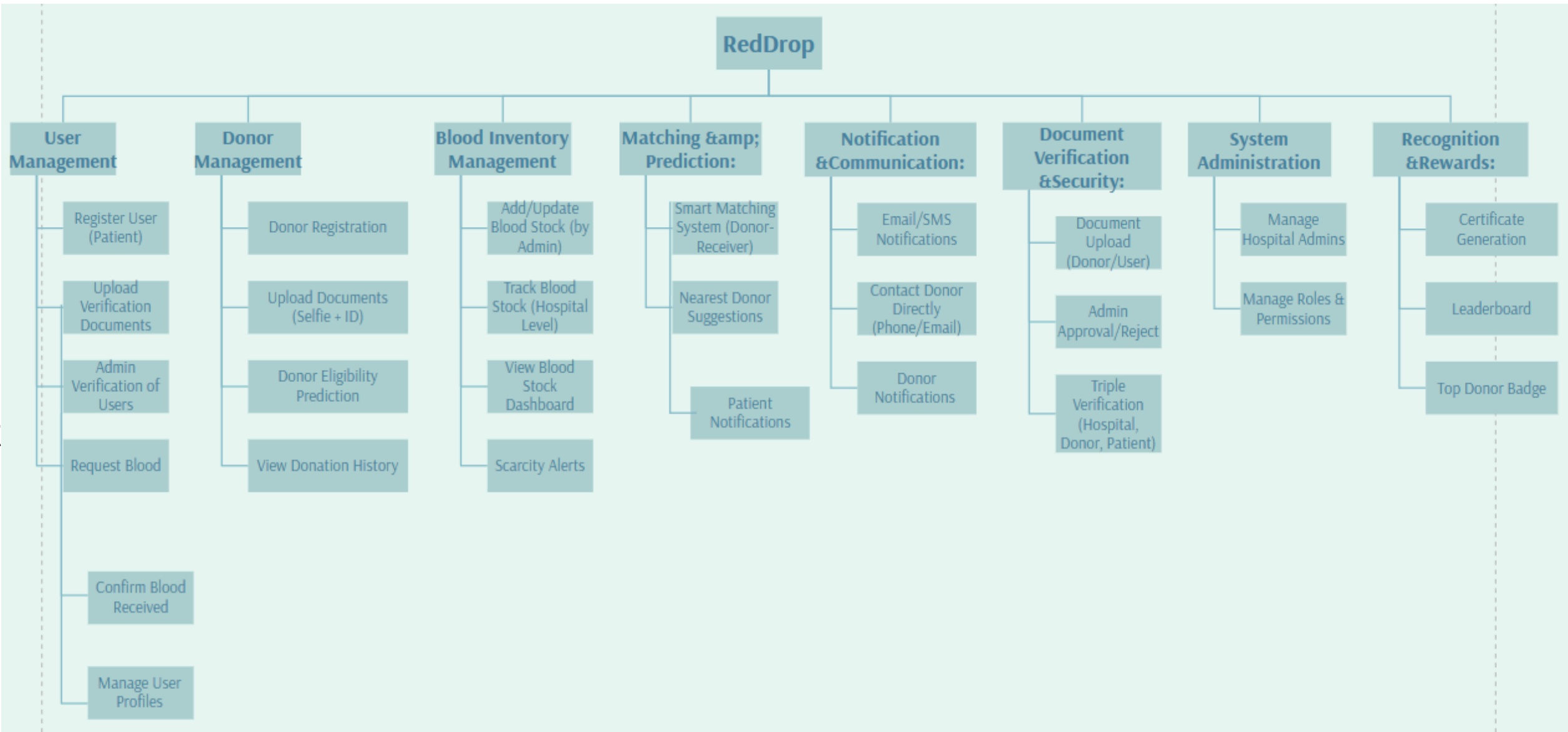
Testing Approach

For the verification and validation during development, Unit Testing has is performed in the individual components where all testable parts has been tested.

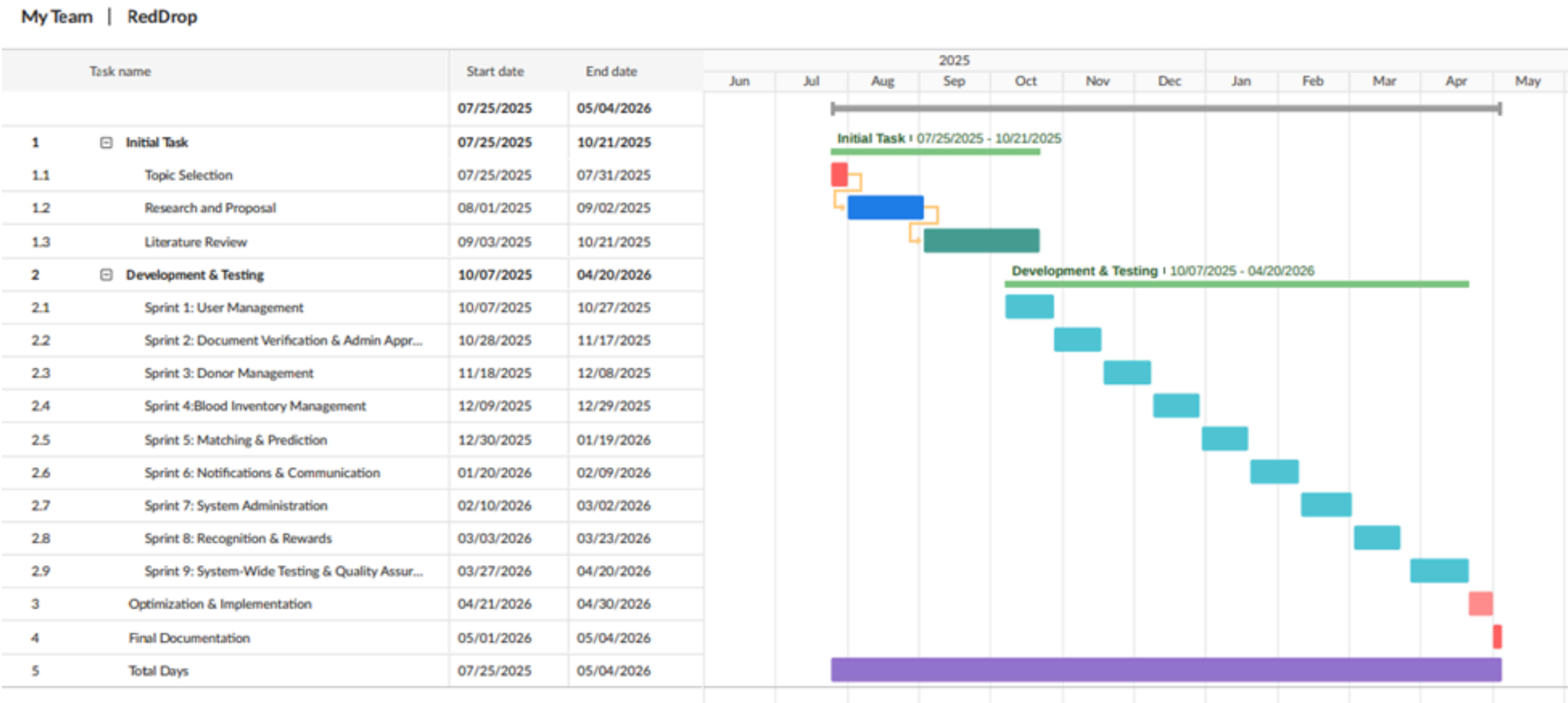
Tools Used

For the frontend, HTML, CSS (Tailwind CSS), and JavaScript are used. The Python Django framework is used for the backend and PostgreSQL for the database.

Functional Decomposition Diagram



Gantt Chart Highlighting Milestones



Research Evidence

Researchers	Focus	Findings	Conclusion
(WORLD HEALTH ORGANIZATION (WHO), 2025)	Blood supply challenges in low- and middle-income countries (LMICs).	LMICs face chronic shortages, inadequate policies, and lack of digitization in blood services.	Highlights the need for digitized systems like RedDrop that provide real-time inventory management and automated notifications.
(Barnes LS, 2022)	Operational inefficiencies in traditional blood collection and transfusion.	Found that 57% of transfusions are delayed due to poor coordination and limited blood stock.	Reinforces RedDrop's goal to improve coordination between patient , hospitals and donors via automated digital workflows.
(Amatya, 2022)	Donor engagement and motivation through recognition.	Recognition systems and gamified features improve donor retention.	Directly informs RedDrop's Rewards Module (certificates, badges, and leaderboards).
HamroLifeBank	Focuses on donor connections and awareness but lacks SMS alerts in emergencies.	Finding the right donor is a time-consuming process for both hospital as well as patients.	RedDrop integrates SMS notification systems to alert donors in emergency situations.

Evaluation and Reflection

- Developed RedDrop blood bank system using Django and PostgreSQL with core features.
- Gained experience in backend development, database handling, and user workflow design.
- Pending features: donor certificate/reward system and stock availability notifications.
- Improved practical knowledge of building real-world applications.